

Olyad Mulugeta

Machine Learning Engineer | Senior Software Engineer

✉ olyadmulugetaa@gmail.com ☎ +251912297301 📍 Addis Ababa, Ethiopia 🔗 LinkedIn 🐙 GitHub

Summary

Machine Learning Engineer and Deputy Director of the Interpretable AI and Natural Language Processing Division at the Ethiopian Artificial Intelligence Institute. Specializes in production ML systems, MLOps, and LLM-powered applications, with a focus on healthcare, NLP, and computer vision. Experienced in building end-to-end ML infrastructure, including data pipelines, model serving, and CI/CD workflows, with an emphasis on low-latency inference and real-world deployment. Led the development and deployment of AI systems for echocardiography-based rheumatic heart disease screening, automated epileptic seizure detection from EEG signals, and wheat leaf disease classification, achieving documented improvements in diagnostic accuracy. Holds a B.Sc. in Software Engineering with a CGPA of 3.7/4.0 and has authored peer-reviewed research on AI-based EEG signal analysis.

PROFESSIONAL EXPERIENCE

Deputy Director, Interpretable AI and Natural Language Processing Division, Ethiopian Artificial Intelligence Institute 09/2023 – Present | Addis Ababa, Ethiopia

- Led the development of an interpretable AI framework for early Rheumatic Heart Disease (RHD) screening, using echocardiographic imaging to improve detection in resource-limited settings.
- Conducted extensive research on epileptic seizure detection using EEG signals, using machine learning and deep learning techniques for seizure classification and prediction for elliptic seizure diagnosis.
- Engineered a deep learning-based wheat leaf disease detection model, achieving 99% accuracy in classifying leaf disease. Implemented Grad-CAM visualization techniques to enhance model interpretability
- Designed and deployed scalable machine learning pipelines, reducing training time by 27%.

EDUCATION

Bachelor of science in Software Engineering, 01/2018 – 07/2023 | Bahir Dar, Ethiopia
Bahir Dar Institute of Technology, Bahir Dar University

- CGPA: 3.7/4.0
- Relevant Coursework: Algorithms and Data Structures, Software Design and Architecture, Database Systems, OOP, Networking, Software Testing and Q, Web Development, Mobile App Development, AI, Machine Learning, Cloud Computing, Cybersecurity
- Projects:
 - Samaritan: An AI-powered application for medicine identification and tracking using image recognition.
 - Enebla: A subscription-based food ordering platform with prepaid orders, personalized recommendations, and automated payments.

SKILLS

Software Engineering

Flutter, Dart, Bloc, Payment Gateways, Python, TypeScript, React, Next.js, Node.js, MongoDB, RESTful APIs, FastAPI

AI & Machine learning

Pytorch, TensorFlow, Pandas, NumPy, Open-CV, Tesseract, YOLO, LLM and Langgraph

MLOps & Infrastructure

Docker, CI/ CD Pipelines, ML Pipelines, GPU Computing, Microservices Architecture, Kubernetes

Data Engineering

Pandas, NumPy, Data Pipelines, Feature Engineering

PROJECTS

Mela Agents, AI-Powered Agentic Platform 02/2025 – 12/2025

- Developed a full-stack web application that generates automated profiles for documents, countries, organizations, and individuals.
- AI-powered speech generation system, enabling automated creation of speeches tailored to different contexts
- Scaled the system for multi-domain applications, supporting organizations in research, media, and government agencies.

CardioLens, Interpretable AI for Echocardiography-Based Early Screening of Rheumatic Heart Disease 01/2022 – 01/2023

- Developed an AI-driven echocardiographic analysis framework, achieving a 70% improvement in diagnostic accuracy for early RHD screening.
- Trained and deployed deep learning models to segment 5,000+ echocardiographic images, enabling precise measurement of mitral regurgitation
- Collaborated with 5+ healthcare professionals and researchers, ensuring clinical relevance and adherence to WHF diagnostic standards.

CortexAI, Automated Epileptic Seizure Detection from EEG Signals Using Image-Based CNNs and Transfer Learning

09/2025 – 01/2026

- Developed an end-to-end seizure detection system using deep learning to classify EEG signals as seizure vs. non-seizure.
- Transformed raw EEG data into time-frequency image representations for CNN-based classification.
- Implemented and optimized an **EfficientNetV2-S** model, outperforming LSTM, 1D-CNN, EEGNet, and Transformer-based approaches.
- Trained on a large-scale dataset (106,800 EEG segments) combined with hospital data; applied fine-tuning for local clinical adaptation.
- Achieved **96.57% accuracy (F1: 96.56%)** on mixed test data and **96.01% accuracy (F1: 96.20%)** on local dataset.
- Built and deployed a full-stack application using **FastAPI (backend)** and **React (frontend)** to serve real-time model predictions.

WheatWise, Deep Learning for Automated Wheat Leaf Disease Detection

01/2024 – 12/2024

- Developed and deployed a fastapi backend system and a mobile application in Flutter, enabling farmers to diagnose wheat diseases via smartphone.
- Compiled a dataset of 50,000+ annotated wheat leaf images, covering 5 common leaf diseases.
- Developed and trained deep learning models, achieving 99.2% classification accuracy, outperforming traditional expert-based diagnosis.

LeshAI, A Deep Learning Approach for Multiclass Detection in Cutaneous Leishmaniasis

09/2025 – 01/2026

- Developed an automated object detection system for diagnosing cutaneous leishmaniasis from microscopic images using the YOLOv11 deep learning model.
- Built and annotated a dataset of 4,000 images (clinical + public sources) with three classes: parasites, healthy monocytes, and infected monocytes.
- Trained and evaluated the model, achieving 76.3% mAP@0.5, with 61.7% precision and 52.2% recall across multi-class detection tasks.
- Designed a multi-class detection pipeline to improve diagnostic interpretability compared to single-class approaches.
- Developed a full-stack application using FastAPI (backend) and Flutter (mobile frontend) for real-time, portable diagnosis.
- Enabled scalable, low-resource deployment to support faster and more consistent diagnosis in underserved regions.

Learn and Transcribe ET Lang V2,

01/2023 – 12/2023

AI-Powered Natural Language Processing Solutions for Ethiopian Languages

- Developed and Deployed a fastapi backend system and a mobile application in Flutter, providing OCR, transcription, voice synthesis, and translation for Amharic and Afan Oromo.
- Integrated speech synthesis with a 95% intelligibility rate, enhancing accessibility for visually impaired users.
- Optimized the system for low-resource devices, reducing latency by 20%, ensuring real-time performance.

EFCD, Ethiopian Food Composition Database

09/2025 – 01/2026

- Served as backend engineer and project manager for a national food database digitization project, designing the core REST API and relational database schema in PostgreSQL.
- Implemented Elastic Search and RAG for autocomplete and fuzzy search.
- Deployed the application using Docker, ensuring high availability and automated backups.

CERTIFICATES

AI Infrastructure and Operations Fundamentals

- Certified in designing, deploying, and optimizing AI infrastructure, ensuring efficient data pipelines for large-scale AI models.
- Gained expertise in GPU-accelerated computing, model deployment, and AI workflow automation

DDD, Hexagonal/Clean Architecture, Tools and Technologies

- Demonstrated proficiency in Domain-Driven Design (DDD), Hexagonal Architecture, and Clean Architecture, enabling the development of scalable and maintainable systems.
- Utilized cutting-edge tools and technologies to implement highly decoupled, flexible, and future-proof architectures across enterprise-grade applications.

Publications

AI-Based Approaches for Analyzing EEG Signals to Identify Epileptic Seizures: A Survey,

24/02/2026

Applied Computational Intelligence and Soft Computing

This paper provides a comprehensive synthesis of AI-based approaches applied to EEG signals for epileptic seizure analysis. We examine 86 peer-reviewed studies from 2014 to 2023 and analyze eleven publicly available EEG datasets, highlighting their characteristics, applications, and implications for model performance.